MILITARY SPECIFICATION

CALIBRATION TECHNICAL BULLETINS AND RELATED ENGINEERING SERVICES

1. SCOPE

1.1 Scope. This specification covers the preparation of calibration technical bulletins, engineering research and evaluation of each item for which a calibration technical bulletin is to be prepared, determination of calibration techniques, and design and fabrication of adapters, special devices which may be required to perform the calibration.

1.2 Classification. The engineering services to be utilized in the preparation of calibration technical bulletins are classified by the following types (see 6.5).

Type I. All requirements of this specification.

Type II. All requirements of this specification less the following engineering services:

3.2 Engineering Report
3.3.4 Accessories
3.3.5 Engineering Drawings
4.4 Accessories
4.5 Drawings
4.6 Engineering Report

Type III. All requirements of this specification less the following engineering services:

3.2 Engineering Report
3.3.4 Accessories
3.3.5 Engineering Drawings
3.3.6 Dynamic Evaluation
4.4 Accessories
4.5 Drawings
4.6 Engineering Report
4.8 Dynamic Evaluation

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, or request for approval, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL–D–70327 — Drawings, Engineering and Associated Lists

STANDARDS

MILITARY

MIL–STD–1243 — Engineering Drawings

PUBLICATIONS

MILITARY

AR 380–5 — Army Regulations Safe-guarding Defense Information

RA NMPC No. 1A — Instructions for Preparation of Manuscripts

FSC–MISC.

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RA NMPC No. 1B	— Instructions for Preparation of Illustrations
RA PUB No. 1C	— Instructions for Typing of Manuscripts
SM 9-4-4931-J38	— Electronic Equipment Standards Set: Secondary Reference (4931-621-7878)
SM 9-4-4931-J39	— Electronic Equipment Standards Set: Secondary Transfer (4931-621-7877)

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The work to be performed under this specification shall be divided into phases, with a summary report and review to be made at the completion of each phase.

3.2 Engineering Report (phase I). The contractor shall prepare an engineering report (phase I) after performing a research and evaluation of each item for which a calibration technical bulletin is to be prepared. The engineering report shall contain the following information:

3.2.1 *Item identification.*

(a) Nomenclature

(b) Federal stock number

(c) Part number

(d) Manufacturer and model number

(e) Weight and size

(f) Reference materiel utilized to prepare the report

3.2.2 *Characteristics.*

(a) Each operating characteristic shall be listed that can be utilized to verify the operation of the item, within the design specification utilized for acceptance inspection.

(b) The difference among models of the item being calibrated.

3.2.3 *Equipment required.* List the DA authorized standards and accessories on separate tables indicating the federal nomenclature, stock number, and the manufacturer's name and model number.

3.2.4 *Additional equipment required.* When standards or accessories are required which have not been authorized as DA standards or accessories, the contractor will be required to submit the information listed below. (If there are no additional standards or accessories required, indicate "None required.")

(a) Noun name of the item.

(b) The name of every known manufacturer and a model number.

(c) The item characteristics.

(d) The approximate cost of the item.

(e) If the item is not commercially available and must be fabricated, an engineering sketch indicating a proposed design with a wiring diagram, shall be submitted with the report.

3.2.5 *Calibration.*

(a) The contractor shall submit a calibration technique to verify each characteristic listed in subparagraph 3.2.2. When more than one technique is applicable, the alternate methods shall be indicated.

(b) Each technique shall be accompanied with a sketch indicating the equipment setup.

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3.2.6 *Submittal.* The contractor shall submit six (6) copies of the engineering report with all source data utilized in the preparation of report, to the procuring activity for approval.

3.2.7 *Approval.* The contractor shall not proceed to the preparation of the preliminary manuscript (phase II) until all portions of the phase I engineering report have been approved by the procuring activity.

3.3 Preliminary manuscrapt (phase II). The contractor shall accomplish the following under the phase II requirement of this specification.

3.3.1 *Preliminary manuscript.* The contractor shall prepare a preliminary procedure in the style and format specified in Appendix I.

3.3.2 *Illustrations.* The contractor shall prepare a halftone illustration of the item being calibrated, and views of all other controls and components that are referenced in the manuscript.

3.3.3 *Line drawings.* The contractor shall include equipment setup diagrams (line drawings) indicating all equipment connections and identifying all accessories.

3.3.4 *Accessories.* If required, the contractor shall fabricate any adapter or special device which has been approved in the phase I engineering report.

3.3.5 *Engineering drawings.* The contractor shall, upon approval of the phase I engineering report, prepare a complete set of engineering drawings for all accessory items to be fabricated and for all commercial items recommended which are not covered by official drawings. The engineering drawings shall be prepared to conform with the requirements of Class 1, Specification MIL-D-70327.

(a) *Types and content of engineering drawings.* Selection of the types and content of engineering drawings shall be governed by Standard MIL-STD-1243, and shall be suitable for unrestricted competitive fixed-price procurement as a complete item.

(b) *Drawing formats and numbers.* Drawing formats and numbers will be supplied by the government.

3.3.6 *Dynamic evaluation.* The contractor shall dynamically evaluate the preliminary manuscript with accessories, at his own facility or at sites specified by the procuring activity. The procuring activity reserves the right to have representatives witness the contractor's dynamic evaluation.

3.3.7 *Additional standards required.* Equipment which the contratcor recommended, and the government has authorized by approval of the phase I engineering report, will be available for the contractor's dynamic evaluation at locations specified by the procuring activity.

3.3.8 *Submittal.* The contractor shall submit the following to the procuring activity for review and evaluation.

(a) Six (6) copies of an illustrated preliminary manuscript with all the source data utilized to prepare the manuscript.

(b) When required, all fabricated or commercial accessory items utilized by the contractor in the dynamic evaluation that are not listed as an authorized accessory item.

(c) When required, a compete set of engineering drawings for all fabricated or commercial accessory items not covered by official drawings.

3.3.9 *Approval.* The contractor shall not proceed to the preparation of the final manu-

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script (phase III), until all portions of phase II have been approved by the procuring activity.

3.4 Final manuscript (phase III). The contractor shall incorporate all changes generated by the review and evaluation of the phase II preliminary manuscript performed by the procuring activity.

3.4.1 *Format and style.* The final manuscript shall be prepared in accordance with the format and style specified in Appendix I, and the applicable requirements outlined in RA NMPC Instruction No. 1A and RA PUB No. 1C.

3.4.2 *Board artwork.* The contractor shall prepare the final board artwork in accordance with RA NMPC Instruction No. 1B.

3.4.3 *Submittal.* The contractor shall submit the following to the procuring activity for final review and approval.

(a) Twelve (12) bound, illustrated copies of each publication with all source data utilized to prepare the manuscript.

(b) Multilith masters of text and illustrations as specified in RA NMPC Instruction No. 1A.

(c) Board artwork for each illustration scaled for a king size publication 7⅞ x 10¼.

(d) One (1) copy negative of each piece of board artwork.

(e) Check list of illustration in duplicate.

(f) Copyright statement in duplicate.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Publication. The contractor shall insure that, prior to the submission of the technical manuscript, the material to be reviewed by the procuring activity is technically adequate, accurate, and prepared in the editorial style and format specified in Appendix I.

4.3 Board artwork. The contractor shall insure that, prior to submission, board artwork shall meet all the requirements specified in RA NMPC Instruction 1B.

4.4 Accessories. The contractor shall insure that, prior to submission of all fabricated accessory items, material submitted shall meet all military specifications and standards.

4.5 Drawings. The contractor shall insure that, prior to the submission of the engineering drawings, the drawings and associated lists shall be prepared in accordance with Specification MIL–D–70327.

4.6 Engineering report. The contractor shall insure that the engineering report constitutes the ultimate effort in research as well as the validity of the technique presented. The contractor shall further insure that all equipment recommended in the report is required, and shall verify all the pertinent characteristics.

4.7 In-process inspection. Representatives of the procuring activity shall be permitted to inspect the items under preparation at the contractor's facilities. This review will be performed periodically at the discretion of the procuring activity, during the contractor's normal working hours.

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4.8 Dynamic evaluation. The contractor shall dynamically evaluate each publication with accessories prior to the submission of the phase II preliminary manuscript. The procuring activity will dynamically evaluate the publication again during the phase II review to insure that the manuscript meets all the requirements specified herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging of all material shall be in accordance with the contractor's best commercial practice ot assure safe delivery.

5.2 Shipping and delivery. All material shall be shipped to the location specified by the contracting officer.

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. The calibration technical bulletins, adapters, and special devices to be prepared, are intended to be used by DA calibration teams and facilities for the calibration of test equipment utilized by Department of Defense.

6.2 Ordering data. Procurement documents shall specify the following.

(a) Title, number, type, and date of this specification.

(b) Delivery schedule.

(c) List of items for which calibration technical bulletins will be prepared.

6.3 Liaison with the government. The contractor is responsible for maintaining adequate liaison with the procuring activity.

6.4 Consultation and assistance by the government. Representatives of the procuring activity will be available for consultation and assistance.

6.5 Engineering Services.

6.5.1 *Type I.* Type I class of this specification will require all the engineering services specified herein.

6.5.2 *Type II.* When type II class of this specification is utilized, the procuring activity must supply the contrator with an engineering report, thereby eliminating the requirement for the phase I engineering services.

6.5.3 *Type III.* When type III class of this specification is utilized, the procuring activity must supply the contractor with a dynamically evaluated manuscript, thereby eliminating the requirement for the phase I engineering services and phase II dynamic evaluation.

6.6 Source Data. The contractor shall submit with each manuscript phase all source data utilized to prepare the manuscript. The procuring activity will return the source data with their comments and approval after performing a review on the manuscript. The contractor shall submit, for the government's retention, all source data utilized in the preparation of the publication with the submittal of the final manuscript.

Custodian:
Army—MU

Preparing activity:
Army—MU
Project No. M103–A002

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APPENDIX I

10. Scope. This appendix to Specification MIL–C–46394 (MU), Calibration Technical Bulletins and Related Engineering Services, provides instructions for the preparation of Department of the Army Calibration Technical Bulletins as required by the basic specification.

20. General Requirements. The calibration technical bulletins prepared under the basic specification shall meet all the quality assurance provisions specified in section 4 of the basic specification, and shall be prepared utilizing the style and format indicated herein.

20.1 The titles and paragraphs listed on title page and utilized in the technical bulletin shall be standardized as shown herein, with the title of each performance check directly referenced to item characteristic being calibrated.

*TB Number/

DEPARTMENT OF THE ARMY TECHNICAL BULLETIN

CALIBRATION PROCEDURE FOR:

(Insert Federal nomenclature and Federal stock number of item to be calibrated.)

Headquarters, Department of the Army, Washington 25, D.C. 1963

SECTION I. GENERAL — Paragraph

Purpose and Scope 1
Reporting of equipment manual improvements 2
Description 3
General instructions 4
Difference among models 5

II. CALIBRATION

Equipment required 6
Preliminary procedure 7

* The publications sequence number and technical bulletin number shall be assigned by Frankford Arsenal. Philadelphia, Pa. 19137.

20.2 The presentation of text outline above shall be mandatory in all calibration technical bulletins prepared for the Department of the Army. The manuscript paragraphs shall be identified as follows:

(a) Main paragraph will utilize a number sequence e.g. 1.

(b) First subparagraph will utilize a letter sequence e.g. *a.*

(c) Second subparagraph will utilize a number in parentheses e.g. (1).

(d) Third subparagraph will utilize a letter in parentheses e.g. (a).

(e) To indicate to the printer where an illustration and its title should be located in the printed copy, a cutline will be inserted between

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paragraphs or subparagraphs immediately following the related text.

20.3 The technical bulletin shall contain the sentence structure outlined in section I below:

Section I. GENERAL

1. Purpose and Scope.

a. This bulletin provides information for the periodic calibration of (insert the Federal nomenclature and stock number) and is to be used by calibration personnel. Since calibration personnel are trained and qualified in the usage of calibration standards, detailed instructions concerning the operation and use of the standards are not contained in this bulletin.

b. Integrated within this bulletin are illustrations delineating the location of all controls and components utilized in this calibration procedure as well as diagrams showing equipment setups. Equipment ground connections are not necessarily shown in the diagrams.

2. Reporting of Equipment Manual Improvements. The direct reporting of errors, omissions and recommendations for improving this equipment manual by the individual user, is authorized and encouraged. DA Form 2028 (Recommended Changes to DA Technical Manual Parts Lists or Supply Manual 7, 8 or 9) will be used for reporting these improvements. This form may be completed using pencil, pen or typewriter. DA Form 2028 will be completed in triplicate and forwarded by the individual using the manual. The original and one copy will be forwarded direct to: Commanding Officer, Frankford Arsenal, ATTN: SMUFA 3200, Philadelphia, Pa., 19137.

3. Description. (This paragraph shall contain a brief and precise description of the instrument and its functional use.) The last sentence in this paragraph should indicate the following: Additional data is listed in *a* through *c* below.

a. Identification

Nomenclature _____
Federal stock number _____
Part number _____
Manufacturer _____
Manufacturer's model number _____
Size _____
Weight _____
Reference _____
(List any material utilized which will provide information to support data in the manuscript such as DA Technical Manual, Manufacturer's Manual, etc.)

b. Specification. (Subparagraph *b* shall be titled Specifications and list the following information:)

(1) For standard type equipment (commercial) manufacturer's specification shall be listed which are normally found in the manufacturer's manual supplied with the equipment.

(2) For special type equipment (peculiar) the specifications listed in the military purchase description shall be cited.

Subparagraph *c* shall be titled *Program Data* and list the following information:

Calibration interval (in days).

Time required for calibration (in hours approx.).

Calibration level (Secondary transfer, Reference, or Primary).

Calibration technique (Low frequency, Physical, or Microwave).

4. General Instructions. *a. Calibration*

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Data Card. During the performance of this procedure annotate DA Form 2382 (Calibration Data Card).

b. Unit Under Test. (This subparagraph shall make first reference of the item being calibrated as the "unit under test.")

5. Difference Among Models. This paragraph shall indicate one of the two conditions listed in *a* and *b* below:

a. None.

b. Cite all models to which the procedure is applicable.

Section II. CALIBRATION

6. Equipment Required. (Equipment required for the performance checks and adjustment shall be listed in table I. Table IA shall list the authorized standards as exemplified below:)

TABLE I. *Equipment required for calibration performance checks and adjustments*

A. Authorized Standards

Nomenclature	Federal stock No./ identifying No.	Manufacturer and model No.
VOLTMETER, ELECTRONIC	6625-555-9810 (8616477)	Hewlett-Packard, model 410B

(Table IB shall list the authorized accessories as exemplified below.)

B. Authorized Accessories

Nomenclature	Federal stock No./ identifying No.	Description
CABLE ASSEMBLY, RADIO FREQUENCY	4931-843-2792 (7907467)	36-in. RG-58/U with BNC plug terminations

(Table IC shall list any additional calibrated equipment or accessories required for the procedure as exemplified below.)

C. Additional Equipment Required

Nomenclature	Federal stock No./ identifying No.	Manufacturer and/or model No.
DUMMY LOAD, ELECTRICAL INVERTER	8901925	Martin Company

Note. Table IC shall only be utilized after it has been determined that it is in the best interest of the Government to utilize calibrated system equipment and accessory items in lieu of standards equipment.

7. Preliminary Procedures. *a.* This paragraph shall contain the necessary instructions required to prepare the unit under test for calibration.

b. The following note shall be inserted as part of paragraph 6:

Note. It is recommended that personnel familiarize themselves with the entire procedure prior to performing calibration.

c. The following note shall be inserted after the last sentence in paragraph 7.

Note. The following paragraphs are divided into subparagraph *a, performance check,* and subpara-

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graph *b*, *adjustments*. When the performance check is within tolerance, do not perform the corresponding adjustment. When the performance check is not within tolerance, perform the corresponding adjustment before continuing with the calibration procedure. When the performance check is not within tolerance and no adjustment is specified, the deficiency must be corrected before continuing with the procedure.

8. (Title of First Check). *a. Performance Check.*

((1) Each paragraph relevant to a verified specification (par. 3*b*.) shall be titled to correlate and identify these specifications. Other checks pertinent to the calibration of the item being calibrated shall also be performed such as internal power supply checks.)

((2) The following sentence shall be utilized when applicable to inform personnel to apply power to the equipment setup "Energize and allow sufficient time (indicate time required) for equipment to warmup and stabilize.")

(3) When stating a specific value to be observed on an instrument, the smallest readable division should be indicated with an additional estimated figure given if practical. The estimated figure should be indicated by a dash above the last digit and the following note shall be used:

Note. A dash above a number indicates a ± 1 deviation is acceptable.

((4) Values indicated in performance checks shall be given as minimum and maximum figures. When tolerances cannot be determined after investigation, either a minimum or maximum value may be cited. The words "exact" or "exactly" shall not be used. This type of value, when required, shall be given as a value without tolerances.)

((5) In all procedures, individual performance checks and adjustments shall be arranged in a logical sequence to keep changes in connections, equipment setups, and control positions to a minimum. In other words, the respective section should be smoothly continuous in nature and should assume that control positions and connections made in preceding paragraphs still exist in subsequent paragraphs unless they are specifically changed. For example, assume that in paragraph 8, a range switch was set to the X10 position and no changes were made to this switch position through the remainder of paragraph 8, the next paragraph can be written with the knowledge that this switch is still in the X10 position. A similar concept shall be applied to the equipment connections and hookups.)

((6) All readings cited in performance checks and adjustments shall be derived from a consideration of the number of significant figures and resolution of the particular instrument involved.)

((7) The last step in each performance check shall be as follows:

Annotate block 10, _____ on DA Form 2382. (The blank will be filled in at the time of dynamic evaluation.) (See 3.3.6)

b. Adjustments.

((1) When adjustments to the item being calibrated are not possible the following statement shall be inserted: "No adjustments can be made.")

((2) Adjustments shall be made to nominal values, when practical, following a logical sequence of equipment operation. Adjustments of an unusual or very difficult nature will be preceded by a note containing special instructions and information as required.)

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((3) When applicable, a correction chart or graph shall be prepared to correct deficiencies.)

9. Illustrations. *a.* The technical bulletin shall contain a halftone illustration showing the front panel view of the unit under test with those controls and switches called out that are mentioned in the text portion of the procedure.

b. If required other halftone illustrations will be included to show other controls and components that are to be adjusted.

c. Equipment setups will be shown using line drawings which shall clearly indicate equipment connections.

d. Final determination of illustrations required will be made at the time of dynamic evaluation.

e. All illustrations prepared to support the technical bulletin shall be prepared in accordance with RA NMPC Instructions No. 1B.

10. General Concepts.

a. Personnel trained in the use of calibration standards will be utilizing Department of the Army calibration procedures. Therefore, instructions for the operation of applicable standards shall not be included in the technical bulletin.

b. The performance check portions of the technical bulletin shall provide detailed instructions for determining the condition of the unit under test as indicated in (1) through (3) below.

(1) Already functioning within specified limits.

(2) Operable but requiring adjustment.

(3) Not in condition to be calibrated until the deficiency has been corrected.

c. The adjustment paragraphs of the technical bulletin shall provide detailed instructions for the adjustment of the unit under test when possible.

11. Technical Policies.

a. Tolerances will be stated showing upper and lower limits where practical.

Example: "48 to 52 volts DC"

b. Associated system test equipment may be utilized in the calibration procedure, only when this equipment is normally used in conjunction with the unit under test. This associated equipment shall be calibrated prior to being used for calibration purposes. It shall be listed in Table 1C "Additional equipment required," with the footnote indicating that it is direct support unit (DSU) equipment.

c. Unless required, calibration shall be conducted without connection of equipment to internal components of the unit under test.

d. Peculiarities of special type test equipment may deem it more feasible to calibrate some special type test equipment in its system environment rather than in a calibration facility or van. Determination of the location area for the calibration shall be coordinated with the procuring activity during the development of the procedure.

e. Disconnection of soldered components or components that could damage the equipment being calibrated through constant disconnecting should not be reflected in technical bulletins unless required.

f. Standard calibration techniques approved by the Army Metrology and Calibration Center of Frankford Arsenal shall be utilized in procedure when possible.

g. Only the computed figures, derived from a mathematical formula shall be listed in the procedure. The formula used to derive the resultant values shall not be listed.

12. Editorial Policies and Standard Terminology.

a. When reference is made to a standard or an accessory for the first time in technical bulletins, both the noun name and Federal stock number shall be used. If the Federal stock number is not available an Army part

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number shall be used. Thereafter only the noun name shall be utilized except when there is more than one item with the same noun name. In this case, the noun name and the Federal stock number shall be used throughout the procedure.

b. In illustrations showing equipment set-ups, the Federal stock number, if available, shall accompany the noun name each time it is used.

c. The unit being calibrated shall be referred to by nomenclature and Federal stock number until paragraph 4*b*, thereafter it will be referred to as "unit under test."

d. When describing the action desired of a toggle switch, the verb "set" shall be used.

Example: Set MOTOR switch to OFF

e. When describing the action of a toggle-type switch that does not have front panel nomenclature designating its position, the verb "set" will be used with the position of the switch denoted

Example: Set power switch to ON (down position).

f. Front panel or internal nomenclature shall always be used, as it appears, in capital letters. If no nomenclature appears, common name in lower case letters will be used.

Example: Set power switch to ON

g. When describing the action desired of a spring-loaded, toggle-type switch that has only one action that can be performed on it, the word "operate" shall be used.

Example: Operate GROUND switch

h. When describing the action of a rotary type switch or dial, the verb "turn" shall be used.

Example: Turn SYNC SELECTOR switch to EXT.

i. When describing the action of a spring-loaded rotary type switch that has only one action that can be performed, the word "operate" shall be used.

Example: Operate RELAY switch

j. When actuating a push button type switch, the verb "press" shall be used.

Example: Press CODER switch

k. When it is necessary to keep a push button type switch in the depressed position, the words "press and hold" shall be used.

Example: Press and hold CODER switch

l. When it is necessary to release a switch, the word "release" shall be used.

Example: Release CODER switch

m. When describing the action of a lever type switch, the verb "set" shall be used.

Example: Set PITCH switch to center position

n. When the action desired is to set the control to a particular position (clockwise or counter clockwise) the verb "turn" shall be used.

Example: Turn AGC control fully counter clockwise

o. When the action desired is for obtaining a result, the verb "adjust" shall be used.

Example: Adjust AGC control for —3.5 volts DC as indicated on AGC METER.

p. When describing the action of a mixed group of controls to be arranged at the same time the word "position" shall be used.

Example: Position controls as indicated in (1) through (3) below:

(1) PULSE RATE CPS control to 500
(2) ATTENUATOR DB switch to 0
(3) PULSE LENGTH μ/sec control fully counter clockwise.

q. When a group of the same type of controls or switches are arranged at one time, use the word applicable to the action required.

r. When describing the appearance of an indicator lamp, use either "indicator glows" or "indicator does not glow."

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s. When describing the action of an indicator lamp glowing or ceasing to glow at a precise time, use the words "glows" or "ceases to glow."

Example: Record time POWER indicator glows

t. Reference to indicator lamps as described in *r* and *s* above shall not include those lamps indicating that an instrument power switch is in the ON or OFF position.

u. When a sentence ends in a plus (+) or minus (—) sign, the words "positive" or "negative" shall follow the sign in parenthesis.

Example: Set TRIGGER slope switch to — (negative)

v. When a symbol appears as panel nomenclature or a part of panel nomenclature the word will be spelled out or abbreviated in capital letters and in parenthesis.

Example:


= indicated in text as (cps)


= indicated in text as (squarewave)

= indicated in text as (OHM)

w. Notes, warnings and cautions, shall be placed within the applicable paragraph and underlined as shown below:

(*Note.* A note will be inserted in the procedure to highlight, an operating procedure, condition, or any fact pertinent to the calibration and should be underlined.)

(*Caution.* A caution note will be inserted in the procedure to highlight any operating procedure, practices, etc., which if not strictly observed will result in damage or destruction of equipment and should be underlined.)

(*Warning.* A warning note will be inserted in the procedure to hightlight any operating procedure, practices, etc., which will result in personnel injury or loss of life if not correctly followed and should be underlined and inserted prior to the operation.)

13. Abbreviations and Examples to be used in tabulated data.

a. 117 v ac ±10%, 50 to 1000 cps, single phase, 350 w

b. 150 v dc ±1%

c. 10 mv/cm to 25 v/cm, ac or dc ±5%

d. DC to 15 mc, usable to 30 mc

e. 10 ranges, 0.01 v/cm to 25 v/cm

f. 1 megohm shunted by 22 mmf

g. 6 steps, 0.1 to 100,000 μsec/cm

h. 2 v peak to peak/cm

i. 10 cps to 750 kc

j. 25 v peak ±5 v

k. 10 pps to 10,000 pps

l. 1.5 μsec ±0.2 μsec

m. 1 kw

n. 5 ma

o. 50 ohms

p. 49.5 dbm maximum

14. Standards for Tables.

a. First column (nomenclature) shall be listed in alphabetical order in tables of authorized standards, accessories, and additional equipment required.

b. The percentage sign (%) shall be used in tables. The word "percent" shall be utilized in text.

c. Vertical and horizontal lines shall separate each entry.

d. Periods shall be used in tabular listing when the entry is a complete statement.

15. Standards for Text.

a. The word "The" shall not be used except in cases where clarification is necessary.

b. The word "position" shall not be used with front panel nomenclature.

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c. Figure and paragraph references shall be abbreviated when placed in parenthesis, and spelled out in all other cases.

Example: *a.* Refer to test equipment setup (fig. 37)
b. Refer to figure 37.

d. Abbreviations shall be used for measurements in the text unless the abbreviation is only one letter, e.g., volt, watt, etc.

16. Procedure Levels. Technical bulletins shall be written to reflect the three levels of calibration, (1) Secondary transfer, (2) Secondary reference, and (3) Primary reference.

a. Secondary Transfer procedure

(1) In secondary transfer procedures when the standard to the unit under test accuracy ratio is 3 : 1 or greater, the standard error should be ignored and the accuracy of the unit under test should be quoted. When the ratio is less than 3 : 1 the square root formula for the sum of the squares shall be utilized to compute the calibration accuracy.

Example of square root of sum of squares
Accuracy of standard = 1%
Accuracy of unit under test = 2%
Formula: $\sqrt{1^2 + 2^2} = \sqrt{5}$
Answer: 2.2 Calibration accuracy

(2) Secondary transfer procedures shall include such checks as internal power supplies and other checks that insure a reliable and calibrated piece of equipment.

b. Secondary Reference procedures

(1) Secondary reference procedures shall reflect the additive errors of the measurement system when the calibration accuracy is being determined.

(2) Secondary reference procedures shall utilize the latest laboratory measurement techniques and reflect the highest laboratory standards. Fixed internal power supplies and checks which are not pertinent to the calibration shall not be reflected in the procedures.

c. Primary Reference procedures

(1) Primary reference procedures shall not be specific and shall be written so that they may be utilized for the calibration *of any* instruments which have the same design characteristics and specifications.

(2) When primary certificates are required, detailed instructions pertaining to their preparation shall be contained in the procedure.

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